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A molecular beam investigation of the catalytic oxidation of CO on Pd (111)

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A detailed investigation of the steady-state and nonsteady-state reaction $\text{CO} + 1/2\text{O}_2 \rightarrow \text{CO}_2$ on Pd (111) has been carried out with the molecular beam technique. It could be shown conclusively that the reaction proceeds between two adsorbed species (Langmuir-Hinshelwood mechanism) throughout the temperature and pressure range investigated. For low CO coverages, the activation energy of the reaction was determined to be 25 kcal/mole, whereas at moderate CO coverages, a rearrangement of the oxygen adlayer takes place resulting in a reduction of the activation energy to 14 kcal/mole. It is not possible to formulate a simple kinetic expression for the reaction rate which is valid over the entire range of temperatures and pressures due to changes in the adsorption rate for O_2 , coadsorption of CO and O_2 , to diffusion in the adlayer, and to changes in the geometrical arrangement within the adlayer with varying coverage.

I. INTRODUCTION

The oxidation of carbon monoxide catalyzed by platinum metals is of particular practical importance, and is on the other hand considered as obeying a relatively simple microscopic reaction mechanism. Numerous studies on this reaction with Pt,¹ Ir,² Ru,³ Rh,⁴ and particularly also Pd⁵⁻²⁴ catalysts have been performed in the past, without however uniquely clarifying the problem of the reaction mechanism. There is general agreement that dissociative chemisorption of oxygen is a necessary prerequisite for the reaction as well as that preadsorbed carbon monoxide acts as an inhibitor for O_2 adsorption and therefore for the reaction itself. The essential question, however, is whether CO_2 formation takes place by interaction between both reactants in the chemisorbed state (Langmuir-Hinshelwood mechanism) or by a collisiontype mechanism with CO striking the O adlayer from the gas phase or eventually weakly held in a physisorbed state (Eley-Rideal mechanism). Whereas this latter description was favored in earlier work (mainly based on arguments concerning the steady-state reaction kinetics),¹⁴ it became soon evident from nonstationary measurements that the LH mechanism definitely may take place and plays at least the role of a parallel path under steady-state conditions.¹⁸⁻²³ Later it was suggested that this reaction catalyzed by Pt¹ or Ir²⁴ may be described exclusively on the basis of the LH mechanism.

It is a well-known fact that a clear decision on the mechanism of a reaction can usually not be made on the basis of the rate laws governing the steady-state kinetics. Instead an investigation of the reaction intermediates (i.e., the surface phases formed) and/or of the time-resolved relaxation behavior is necessary. In a preceding paper,²⁴ the coadsorption and mutual interaction as well as transient product formation of carbon monoxide and oxygen on a Pd(111) surface was studied at low temperatures. The results showed that below 400 K there is in fact no evidence for the operation of the ER

mechanism. However, the adsorption properties of the reactants and thereby also their reactivity may be mutually affected in a rather complicated manner (which was analyzed in detail), so that no appropriate description of the steady-state kinetics of CO_2 formation may be achieved in terms of simple rate laws over the whole range of accessible surface concentrations. The steady-state reaction proceeds with appreciable rates however only above 400 K where the CO coverage is usually fairly low and the same concepts may not necessarily hold as at low temperatures. The fairly rapid relaxation behavior may then no longer be followed by the conventional methods of surface research. It will, however, be demonstrated that the molecular beam technique as applied in the present work is an excellent tool for deciding between different possible reaction paths. The results show clearly that the reaction proceeds via the LH mechanism and no evidence is found for the operation of the ER mechanism even at temperatures applied in steady-state catalysis. In addition, the rate constant for the surface reaction is derived and the overall results were found to fit nicely with the general picture established in the previous low-temperature work.²⁴

Since the steady-state kinetics of CO_2 formation is practically independent of the surface orientation of the Pd catalyst,¹⁷ the present conclusions reached for the (111) plane may safely be transferred to the behavior of other crystal planes. Presumably the general conclusions concerning the reaction mechanism also hold for the other platinum metals which exhibit qualitatively similar reaction kinetics.

II. EXPERIMENTAL

The apparatus used in this work has been described in detail elsewhere.²⁵ Briefly, it consists of separate uhv beam generating and investigation systems both of which are bakeable to 250 °C. The beam generating system is three stage differentially pumped with liquid nitrogen baffled oil diffusion pumps. The nozzle beam enters the investigation chamber through a straight-through valve and can be mechanically chopped before impinging on the substrate. A maximum intensity equivalent to the

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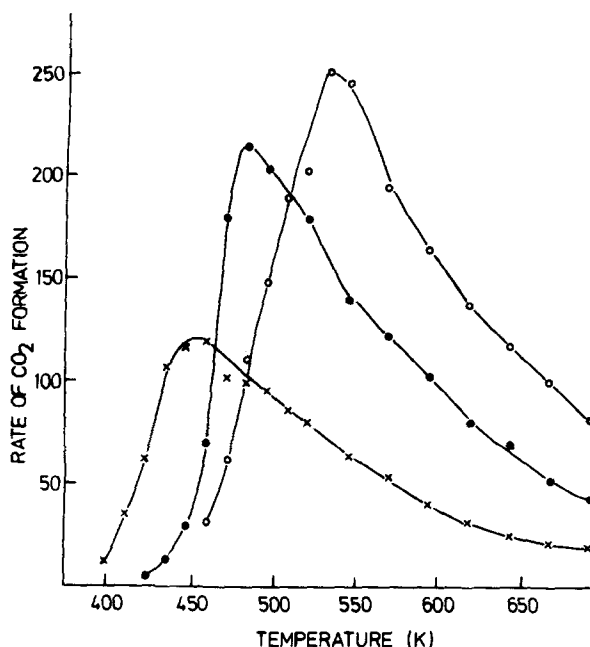


FIG. 1. Steady-state rate of CO_2 production as a function of substrate temperature for an O_2 beam (equivalent pressure 4×10^{-7} torr) and CO pressures of 1×10^{-7} (x), 3×10^{-7} (●), and 1×10^{-6} (○) torr.

arrival rate at a pressure of 4×10^{-7} torr is attained at the substrate (nozzle-substrate distance 1200 mm). The scattered beam was detected with a rotatable quadrupole mass spectrometer which allowed measurements of in-plane angular scattering distributions. The pumping speed in the investigation chamber is increased through titanium sublimations so that the total pressure with the beam on remains below 1×10^{-10} torr.

The results reported in Secs. III. A–III. C were obtained at a fixed scattering angle which was chosen to maximize the intensity of the CO_2 signal after it was established that CO_2 is emitted from the surface in a cosine distribution (see Sec. III. D). Previous work²⁵ has shown that the sticking coefficients for O_2 and CO on Pd(111) are independent of the angle of incidence.

III. RESULTS

A. Steady-state kinetics of CO_2 formation

The steady-state kinetics of CO_2 formation on different single crystal planes was already studied in detail in the past^{14–18} and therefore only relatively few measurements of this type were performed in the present work. These were done by using an O_2 beam of constant intensity and varying CO background pressures, but the results are not affected by the fact that the O_2 molecules strike the surface from a well-defined direction. Since the pumping speed of the vacuum system of the reaction chamber is much larger than the rate of CO_2 formation the partial pressure of CO_2 as measured by means of the quadrupole mass spectrometer is directly proportional to this rate.

Figure 1 shows the variation of the stationary rate of CO_2 formation, $r = d\text{CO}_2/dt$, with temperature at a constant oxygen pressure, $p_{\text{O}_2} = 4 \times 10^{-7}$ torr, and at three

different CO pressures, $p_{\text{CO}} = 1 \times 10^{-7}$, 3×10^{-7} , and 1×10^{-6} torr, respectively. The relatively complex nature of these data agrees with that found in earlier studies and will be explained in the discussion on the basis of the reaction mechanism as supported by the relaxation type measurements.

The dependence of $d\text{CO}_2/dt$ on p_{O_2} at two constant substrate temperatures is shown in Fig. 2 for different CO pressures. Comparison with Fig. 1 reveals that in Fig. 2(a) Curve a ($p_{\text{CO}} = 1 \times 10^{-7}$ torr) corresponds to a situation where the maximum in the $r(T)$ curve is below the working temperature of 485 K; for Curve c ($p_{\text{CO}} = 1 \times 10^{-6}$ torr) the maximum lies above this temperature. As a consequence, different rate laws $r = f(p_{\text{O}_2})$ are observed. This effect is connected with the CO coverages under reaction conditions as discussed later. Quite analogous effects occur at the higher substrate temperature of 522 K [Fig. 2(b)].

Similar experiments were performed with respect to the variation of the rate of CO_2 formation as a function of the CO pressure at constant O_2 pressure and temperature, which are not reproduced in detail. As a general result it is noted that under conditions for which the temperature is above the maximum in the corresponding $r(T)$ curves the reaction rate increases linearly with increasing p_{CO} up to a saturation value above which it remains constant. This value is determined by the relative adsorption rates of CO and O_2 . For high CO beam intensities, the oxygen coverage is depleted through reaction, leading to an apparent decrease in the reaction order in CO from one to zero with increasing beam intensity. At temperatures below the maximum in the $r(T)$ curves on the other hand CO acts as an inhibitor and the rate decreases approximately proportional to $1/p_{\text{CO}}$.

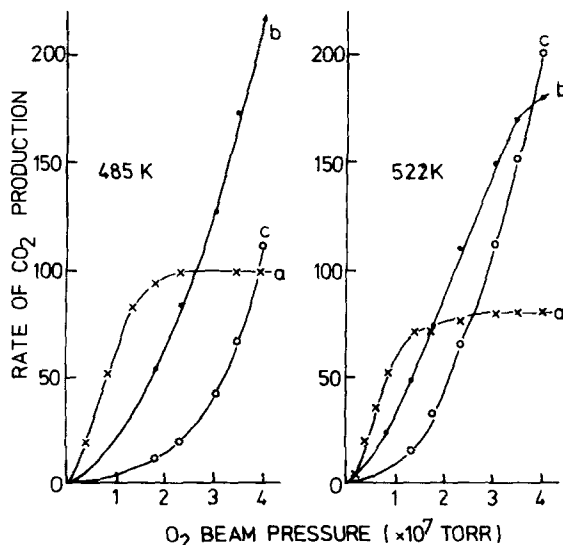


FIG. 2. Steady-state rate of CO_2 production as a function of O_2 beam intensity (equivalent pressures) for various CO pressures. (a) $T = 485$ K, (b) $T = 522$ K. CO pressures: (x) 1×10^{-7} torr; (●) 3×10^{-7} torr; (○) 1×10^{-6} torr.

B. Transient CO₂ formation

The first type of relaxation experiment was performed in the following way: One of the reactants was present with a constant background pressure by direct introduction into the reaction chamber through a variable leak valve. The other formed a molecular beam which was switched on at $t=0$ by suddenly opening a shutter in the beam apparatus, thus producing a steplike pressure rise from zero to a constant value at the surface. This process took place within about 20 msec so that relaxation times exceeding this value could be determined. With some of the experiments the surface was covered with a certain amount of oxygen and subsequently the O₂ atmosphere was pumped off prior to switching on the CO beam. The variation of the CO₂ partial pressure was recorded as a function of time which was again directly proportional to the (transient) rate of CO₂ formation $r(t)$. Similar experiments were performed in previous work²⁴ by producing the pressure "jump" simply by opening the corresponding leak valve, which caused of course a much worse time resolution of at least a few seconds. Nevertheless, valuable comparison with data can be made which were generally performed at lower temperatures than those applied in the present study.

For the measurements illustrated by Fig. 3 the sample was at first exposed (at 300 K) to a constant O₂ partial pressure of $p_{O_2} = 1 \times 10^{-7}$ torr, leading to the buildup of a O-2 $\times$ 2 structure ($\theta_0 = 0.25$).²⁶ Desorption and bulk diffusion as well as the formation of inactive "oxide" species of O₂ can be completely neglected under the applied conditions ($T \leq 700$ K). While the O₂ pressure was kept at its constant value the CO beam was suddenly switched on, giving rise to an effective CO pressure of 6×10^{-8} torr at the sample surface, and the CO₂ partial pressure ($\hat{=}$ rate of CO₂ formation) was followed as a function of time.

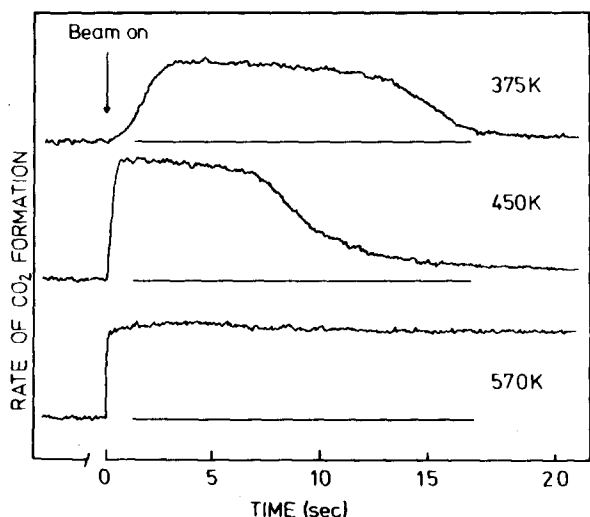


FIG. 3. Non-steady-state rate of CO₂ production as a function of time for a CO beam (equivalent pressure 6×10^{-8} torr) impinging on the substrate for three substrate temperatures. $p_{O_2} = 1 \times 10^{-7}$ torr.

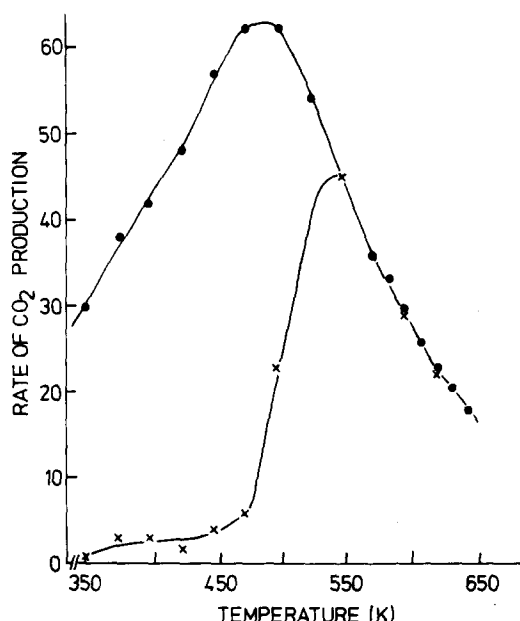


FIG. 4. A comparison of the maximum and the steady state rates of CO₂ production obtained from curves of the type shown in Fig. 3 at various substrate temperatures. $r_{\max}$ ($\bullet$), $r_{\text{steady state}}$ ($\times$).

Three distinct stages can be distinguished with these curves: an initial induction period in which $d^2\text{CO}_2/dt^2 > 0$, a plateau with maximum rate, and for ($T \leq 500$ K) a falloff of the rate to a steady-state value after longer times. The duration of the induction period decreases rapidly with increasing surface temperature but may be identified with a fast storage oscilloscope at all temperatures investigated (< 600 K). The rate at the plateau ($r_{\max}$) increases with increasing temperature and exhibits a maximum at about 500 K, after which it decreases. The steady-state rate (r_{stat}) also increases with increasing temperature and reaches its maximum value at 550 K, after which $r_{\max}$ and r_{stat} are identical. The results are summarized in Fig. 4. The steady-state rate follows a temperature dependence which is of course identical to the data obtained according to the procedure described in Sec. III. A.

Since the maximum rate $r_{\max}$ occurs after different CO exposures for varying temperatures, the curve $r_{\max}(T)$ in Fig. 4 describes the temperature dependence of the rate at considerably differing coverages θ_0 and θ_{CO} . This can be corrected for in the following way: A set of curves similar to those shown in Fig. 3 was recorded with the O₂ pumped off after completion of the O-2 $\times$ 2 structure ($\theta_0 = 0.25$) and prior to CO admission. The rate of CO₂ production was then determined at all temperatures at a time t' such that the relative decrease of the oxygen coverage is still negligible, i.e., $[\theta_0(t) - \theta_0(0)]/\theta_0(0) \ll 1$. The resulting data are reproduced in Fig. 5 corresponding to $\theta_0 = 0.25$ and to t' equivalent to a CO exposure of 0.06 monolayers.

The rate of CO₂ formation for $T < 500$ K is now smaller than the corresponding $r_{\max}$, since in these cases t' lies within the induction period. Under the same conditions the intensity of CO molecules backscattered from

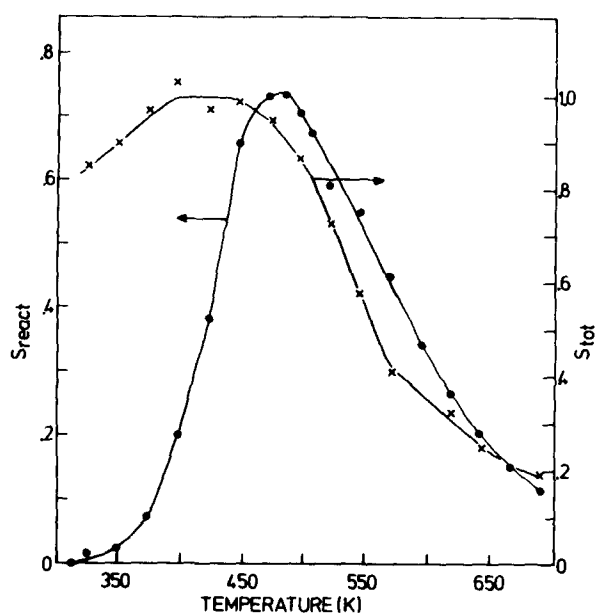


FIG. 5. Reactive sticking coefficient (●) and the total sticking coefficient for CO (x) as a function of substrate temperature. $\theta_0 = 0.25$ and the CO exposure is equivalent to 0.06 monolayers.

the surface was determined. Since the angular distribution of scattered CO is not affected by the coverage, this quantity may easily be converted into an absolute effective sticking coefficient.²⁵ This sticking coefficient, S_{tot} , can be divided into

$$S_{\text{tot}} = S_{\text{react}} + S_{\text{ad}}, \quad (1)$$

in which S_{react} is the probability that an impinging CO molecule is consumed by the reaction to CO_2 , and S_{ad} is the probability for (nonreactive) adsorption. Since one CO molecule disappears for each CO_2 molecule formed, $d\text{CO}_2/dt$ is proportional to S_{react} and the two curves in Fig. 5 yield complementary information. At 350 K, S_{react} is nearly zero and $S_{\text{tot}} = S_{\text{ad}} = 0.8$. The CO molecules are adsorbed on the oxygen saturated surface with a probability which is only about 20% less than that for adsorption on a clean Pd(111) surface ($S_0 = 0.96$ ²⁵). The increase in $d\text{CO}_2/dt$ for 350 K < T < 450 K causes S_{tot} to increase to a value of unity and also to an increase in S_{react} at the expense of S_{ad} . For $T > 500$ K both S_{tot} and S_{react} decrease with increasing temperature. S_{ad} drops sharply above 475 K because the equilibrium coverage at the corresponding CO pressure falls below $\theta_{\text{CO}} = 0.06$ at approximately this temperature. S_{react} decreases for similar reasons: With increasing temperature an adsorbed CO molecule has an increasingly higher chance for desorption relative to the probability for recombination with O_{ad} during its residence time on the surface.

The (momentary) coverages θ_0 and θ_{CO} at time t' underlying the data of Fig. 6 were derived in the following way (assuming a reaction mechanism $\text{O}_{\text{ad}} + \text{CO}_{\text{ad}} \rightarrow \text{CO}_2$, which will be justified during this paper): For each CO_2 molecule formed, an oxygen atom is removed from the adlayer so that

$$\theta_0(t) = \left\{ 1 - \int_0^t \left(\frac{d\text{CO}_2}{dt} \right) dt / \left[\int_0^\infty \left(\frac{d\text{CO}_2}{dt} \right) dt \right] \right\} \theta_{0,\text{max}}, \quad (2)$$

where $\theta_{0,\text{max}} = 0.25$. It was verified experimentally that $\int_0^\infty (d\text{CO}_2/dt) dt$ is independent of the substrate temperature in the range investigated ($T < 700$ K), e.g., eventual losses of adsorbed oxygen by bulk diffusion or O_2 desorption are negligible.

The CO coverage is determined from

$$\theta_{\text{CO}}(t) = I_0 S_{\text{tot}} t - \theta_{0,\text{max}} \left\{ \int_0^t \left(\frac{d\text{CO}_2}{dt} \right) dt / \int_0^\infty \left(\frac{d\text{CO}_2}{dt} \right) dt \right\}, \quad (3)$$

where I_0 is the intensity of the impinging CO beam (in monolayers per sec) and S_{tot} is taken from Fig. 5.

An absolute determination of S_{react} can be made since

$$S_{\text{react}} = \frac{S_{\text{tot}} I_0 t - \theta_{\text{CO}}}{S_{\text{tot}} I_0 t} \quad (4)$$

if t is small enough so that S is independent of time. This yields a maximum value for $S_{\text{react}} = 0.73$ at 472 K. Data for other temperatures are plotted in Fig. 5.

In order to obtain some insight into the dependence of the (isothermal) rate of CO_2 production on the surface concentrations of the reactants instantaneous data at two temperatures were analyzed in detail: The surface was at first precovered with O_{ad} to $\theta_0 = 0.25$, then the O_2 atmosphere pumped off and the CO beam with an effective pressure $p_{\text{CO}} = 6 \times 10^{-8}$ torr switched on. The rate of CO_2 formation, $d\text{CO}_2/dt$, is directly proportional to the recorded CO_2 partial pressure. The absolute value of $d\text{CO}_2/dt$ may easily be derived since $\int_0^\infty (d\text{CO}_2/dt) dt = \theta_{0,\text{max}} = 0.25$. The instantaneous coverages θ_0 and θ_{CO} were determined according to Eqs. (2) and (3).

The resulting data for 374 K are reproduced in Fig. 6. Initial impingement of CO on the surface for which θ_0

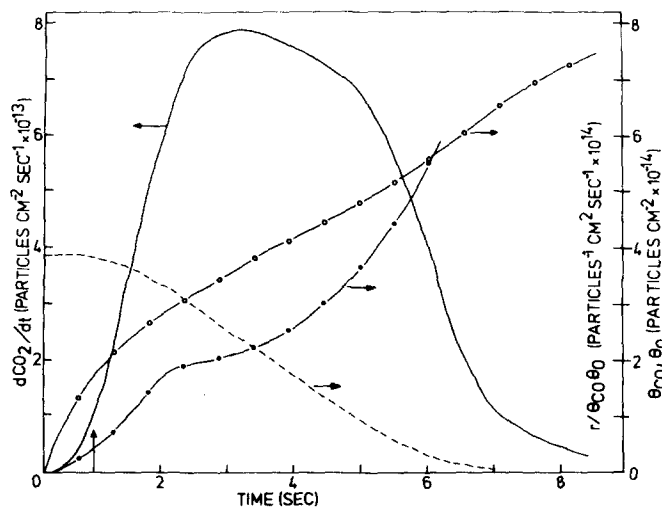


FIG. 6. Rate of CO_2 production, r , (—), θ_0 (---), and θ_{CO} (○) as a function of time for $T = 374$ K. The curve (●) shows $r/\theta_0\theta_{\text{CO}}$ as a function of time.

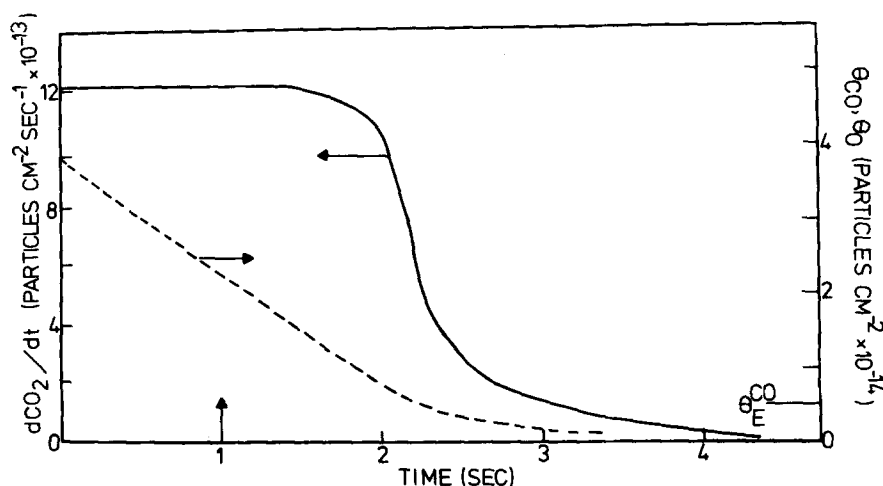


FIG. 7. As in Fig. 6 for $T = 509$ K. Since θ_{CO} cannot be measured directly, only r and θ_{O} are shown as a function of time.

$= 0.25$ leads to incorporation of CO into the adlayer without reaction. The reaction starts with $d^2\text{CO}_2/dt > 0$ and reaches a maximum rate after about 3 sec where $\theta_{\text{CO}} \approx 0.25$ and $\theta_{\text{O}} \approx 0.15$. After 5 sec the rate drops sharply due to consumption of O_{ad} .

The data for 509 K are shown in Fig. 7. The induction period is not resolved with the displayed time scale. θ_{CO} is too small to be directly detectable, but the upper limit is the equilibrium coverage at the respective CO pressure and sample temperature, $\theta_{\text{CO},\text{eq}} = 0.03$.²⁷ It is seen that $d\text{CO}_2/dt$ is independent of θ_{O} for $\theta_{\text{O}} > 0.08$, i.e., the kinetics follows a reaction order of zero with respect to the oxygen surface concentration for $0.08 < \theta_{\text{O}} < 0.25$. Below $\theta_{\text{O}} = 0.08$ the rate turns out to be approximately proportional to θ_{O} (i.e., first order). Independent measurements showed that at this temperature the rate is directly proportional to the CO pressure over the whole range of oxygen coverages. This becomes plausible since under these conditions θ_{CO} is very small and therefore proportional to p_{CO} .²⁷

This discussion shows that a rate law formulated on the basis of the elementary step $\text{CO}_{\text{ad}} + \text{O}_{\text{ad}} \rightarrow \text{CO}_2$, viz.,

$$r = d\text{CO}_2/dt = k' \theta_{\text{O}} \theta_{\text{CO}}, \quad (5)$$

will hardly yield an appropriate description of the reaction kinetics. The thus defined apparent rate constant k' was evaluated from the data at 374 K and is also plotted in Fig. 6. Obviously k' is not a real constant (at fixed temperature) but varies considerably with the surface concentrations of the reacting species. This result is not so surprising if the information on the various possibilities of the mutual surface configurations of O_{ad} and CO_{ad} as investigated in detail in Ref. 24 are taken into account. As a consequence a plot of $\ln k'$ vs $1/T$ will not yield the true activation energy for the LH reaction. The value of 7 kcal/mole which was derived earlier on the assumption of a coverage-independent k' will certainly not be the true activation energy $E_{\text{LH}}^\ddagger$ in view of the present results.

This quantity may, however, be evaluated with a fair accuracy from the data of Fig. 5 for the following rea-

sons: The reaction rate may obviously generally be written as

$$r = kf(\theta_{\text{O}}, \theta_{\text{CO}}), \quad (6)$$

where k is the true rate constant (under the corresponding conditions of surface arrangement of the adsorbates). The data of Fig. 5 were recorded under conditions where θ_{O} varies only by about 20% from $\theta_{\text{O}} = 0.25$ with differing temperature, so that it may approximately be regarded as being constant. For $T < 422$ K, θ_{CO} also remains constant (~ 0.06) to within 15%, and therefore a plot of $\ln r$ vs $1/T$ (at roughly constant θ_{O} and θ_{CO}) should yield the true activation energy for k if $T < 422$ K. (The same result is of course expected if for convenience $(1/\theta_{\text{O}} \theta_{\text{CO}})r$ is used instead of r . This conclusion was explicitly checked.) For $T > 500$ K the rate was found to be proportional to θ_{CO} , whereas θ_{O} still remains constant under the chosen conditions. Therefore again a plot of $\ln(r/\theta_{\text{O}} \theta_{\text{CO}})$ vs $1/T$ should yield the true activation energy in this temperature range. Since θ_{CO} is too small to be measured directly for $T > 450$ K, the equilibrium CO coverage at the beam pressure and surface temperature, $\theta_{\text{E}}^{\text{CO}}$, was used to calculate k .

The result is reproduced in Fig. 8. The Arrhenius plot shows two linear portions corresponding to activation energies of 27 ± 2 kcal/mole at high temperatures (and very small CO_{ad} concentrations) and 14 ± 2 kcal/mole at temperatures below 500 K (and $\theta_{\text{CO}} > 0.02$). All data points of the latter range lie within the induction period, suggesting a different microscopic reaction mechanism.

The experiments described up to now in this section can also be carried out with a CO precovered surface upon which an O_2 beam is impinging. Measurements of this type for three substrate temperatures are reproduced in Fig. 9. Under these conditions CO is not irreversibly adsorbed (as was the case with the just described experiments with preadsorbed oxygen), but instead an (reversible) equilibrium coverage is established, depending on CO pressure and temperature. After reaching this equilibrium with a stationary CO pressure of 3×10^{-8} torr the O_2 beam was switched on to an effective partial pressure of 6×10^{-8} torr (while the

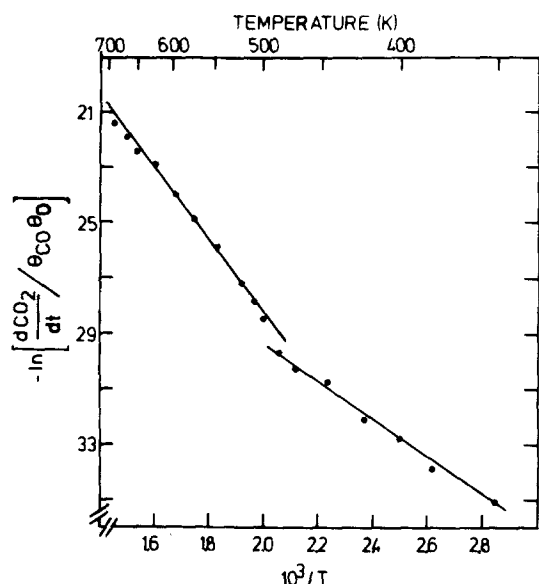


FIG. 8. $\ln k$, where $k = r/\theta_{\text{CO}}\theta_{\text{O}}$, as a function of $1/T$. The data have been obtained from curves of the type shown in Figs. 6 and 7. $\theta_{\text{O}}^{\text{CO}}$ has been used when θ_{CO} cannot be measured directly.

CO atmosphere was kept constant) and the CO_2 partial pressure ($\triangleq d\text{CO}_2/dt$) was followed as a function of time.

The curves of Fig. 9 may qualitatively be easily interpreted: At 415 K, $r = d\text{CO}_2/dt$ is initially zero and slowly approaches the limiting value of the steady-state reaction rate. At this temperature the equilibrium coverage (i.e., without reaction) $\theta_{\text{CO}} \approx 0.35$ at $p_{\text{CO}} = 3 \times 10^{-8}$ torr, and chemisorption of O_2 is strongly inhibited by the CO adlayer.^{16,24} Only after a portion of CO_{ad} has been reacted off can more oxygen be adsorbed on the vacant sites. (At $p_{\text{O}_2} = 2p_{\text{CO}}$ the probability for the formation of an adsorbed oxygen atom is about twice as large as that for CO_{ad} .)²⁵ The slow approach to the steady-state rate is due to the competition of O_2 and CO for empty sites.

At 450 K and $p_{\text{CO}} = 3 \times 10^{-8}$ torr the equilibrium coverage of CO is $\theta_{\text{CO},\text{eq}} = 0.25$ and the sticking coefficient for O_2 is already appreciable (≈ 0.2). The reaction now starts rather quickly and since at these conditions of θ_{CO} , p_{CO} , and p_{O_2} , the rate of adsorption is greater for O_2 than for CO, θ_{CO} decreases, and therefore r passes through a maximum before attaining its steady-state value. The latter was found to increase linearly with increasing p_{CO} until the adsorption rates of CO and O_2 are equal. At 520 K the equilibrium CO concentration is very small so that the behavior is identical to that for a CO beam impinging on an oxygen precovered surface (compare Fig. 3). θ_{CO} is now directly proportional to p_{CO} (at a given O_2 pressure) and a transient maximum of r is no longer observed.

C. Modulated beam experiments

Measurements with a periodically modulated molecular beam allow a time-resolved study of reaction rates.²⁸ A reaction proceeding under steady-state conditions

may be shifted by periodically varying the impingement rates of particles of one of the reactants arriving at the surface, viz., $p_i = p_i^0 + ae^{i\omega t}$, where a is the amplitude and ω the frequency of the modulation. The relaxation time τ reflecting the kinetic response of the reaction system is obtained from the amplitude J and phase lag ϕ of the product signal at the modulation frequency ω . By applying phase-sensitive detection techniques the product signal amplitude A and B may be measured at two phase angles differing by $\pi/2$ from each other. The interesting quantities J and ϕ may then be evaluated through

$$J = (A^2 + B^2)^{1/2} \quad (7)$$

and

$$\phi = \tan^{-1} \frac{B}{A} \quad (8)$$

The dependence of J and ϕ on ω and τ has been calculated for a variety of kinetic processes with differing reaction orders.²⁸ In the case of a first-order (or pseudo-first-order) rate process the rate constant $k = 1/\tau$ is determined by

$$J/J_0 = (1 + \omega^2/k^2)^{-1/2} \quad (9)$$

and

$$\tan \phi = \omega/k, \quad (10)$$

where J_0 is the amplitude observed for $\omega = 0$.

Quantitative data can of course only be obtained if the relaxation times are larger than the experimental detection limit, which was about 2×10^{-5} sec in the present study.

In the present case the modulated beam technique was primarily applied to allow a distinction between Eley-Rideal (ER) and Langmuir-Hinshelwood (LH) types of mechanisms for the surface reaction. The individual reaction steps of these mechanisms as they are pertinent for the present discussion may be formulated as follows:

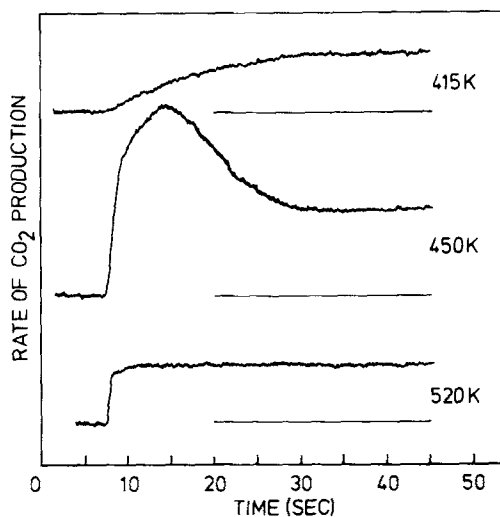


FIG. 9. Non-steady-state rate of CO_2 production as a function of time for an O_2 beam (equivalent pressure 6×10^{-8} torr) impinging on the substrate. $p_{\text{CO}} = 3 \times 10^{-8}$ torr.

(i) Langmuir-Hinshelwood:



(ii) Eley-Rideal:



These simplified equations of course take no account of the complications due to the surface arrangement of the adsorbates and due to the influence of the partial coverages on the sticking coefficients etc. As was shown in the preceding sections, the rate laws governing the reaction kinetics will in general not follow simple first-order expressions as one could expect from an inspection of Eqs. (11) and (12), but this question has to be checked individually for the corresponding experimental conditions.

In the above formulation possible desorption as well as bulk diffusion of oxygen have been omitted, since these steps are negligible for $T < 700$ K. There is furthermore no indication for a reaction of the type $2\text{CO}_{\text{ad}} + \text{O}_2 \rightarrow 2\text{CO}_2$, for obvious reasons. With the ER mechanism no distinction is made between gaseous CO and an eventually physisorbed species (on top of O_{ad}) since under the applied conditions the surface residence time of the latter would be below the limits of detection ($\sim 2 \times 10^{-5}$ sec) and its surface concentration would be directly proportional to the CO pressure. Similar arguments hold for the CO_2 molecules formed.

The following discussion will be concerned with experiments performed at $T > 500$ K where the stationary CO coverage is rather small. As has been outlined in Sec. III. A, under these conditions the rate of CO_2 formation, r , varies linearly with the CO pressure (as long as the stationary oxygen coverage is high enough so that the formation of O_{ad} does not become rate limiting). The rate laws may then be written as

$$\frac{d\text{CO}_2}{dt} = k_4 \theta_{\text{CO}} f(\theta_{\text{O}}) \quad (\text{LH}), \quad (13)$$

and

$$\frac{d\text{CO}_2}{dt} = k_5 p_{\text{CO}} f(\theta_{\text{O}}) \quad (\text{ER}). \quad (14)$$

The reaction order with respect to the oxygen coverage is not simple as was shown above. If, however, the experimental conditions are chosen in a way that θ_{O} is high enough so that $d\text{CO}_2/dt$ is independent of small fluctuations in θ_{O} during the periodic modulation of the CO beam (the O_2 background pressure is kept constant), then $f(\theta_{\text{O}}) \approx \text{const}$ and Eqs. (13) and (14) simplify to pseudo-first-order kinetics. As can be seen from Fig. 7 this requires that $\theta_{\text{O}}(t) > 0.1$, which may easily be fulfilled.

p_{CO} is modulated with a frequency ω so that

$$p_{\text{CO}} = p_{\text{CO}}^0 + \alpha e^{i\omega t}, \quad (15)$$

and

$$\theta_{\text{CO}} = \theta_{\text{CO}}^0 + \beta e^{i\omega t}. \quad (16)$$

Insertion of these expressions into Eqs. (13) and (14) yields for the LH mechanism

$$\frac{d\text{CO}_2}{dt} = k_4 f(\theta_{\text{O}}) \left(\theta_{\text{CO}}^0 + \frac{k_1 \alpha e^{i(\omega t - \phi)}}{[k_2 + k_4 f(\theta_{\text{O}})]^2 + \omega^2} \right) \quad (17)$$

with

$$\tan \phi = \frac{\omega}{k_2 + k_4 f(\theta_{\text{O}})}. \quad (18)$$

The corresponding result for the ER mechanism is

$$\frac{d\text{CO}_2}{dt} = k_5 f(\theta_{\text{O}}) (p_{\text{CO}}^0 + \alpha e^{i\omega t}) \quad (19)$$

with $\tan \phi = 0$.

Thus for the LH mechanism a temperature dependent phase lag for the CO_2 signal is predicted. For $k_2 \gg k_4 f(\theta_{\text{O}})$, which is fulfilled in the temperature range studied (> 500 K), i.e., the desorption of CO_{ad} is fast compared to its reaction with O_{ad} , the temperature dependence of $\tan \phi$ should yield the activation energy for CO desorption. On the other hand, no phase shift should occur at any temperature at which the ER mechanism is dominant. If the ER mechanism is assumed to proceed through a physisorbed CO molecule rather than through a direct collision, $\tan \phi = \omega/k_5 \approx 0$, where k_5 is the desorption rate constant from the physisorbed state. Indeed, a plot of $\ln(\tan \phi / \omega)$ vs $1/T$ yields a straight line (Fig. 10) with a slope equivalent to $E^* = 33$ kcal/mole. This value is within the limits of error identical with the activation energy of CO desorption as determined in independent molecular beam experiments²⁵ (32 kcal/mole) as well as with the isosteric energy of CO adsorption (34 kcal/mole).²⁷ (Since CO adsorption is not activated, the latter two data should be equal.) This result clearly rules out the operation of the ER mechanism for $T > 500$ K and demonstrates that the reaction proceeds via regularly chemisorbed CO molecules under these conditions.

Since in the present case $k_2 \gg k_4 f(\theta_{\text{O}})$, the activation energy for the rate constant, k_4 , of the surface reaction

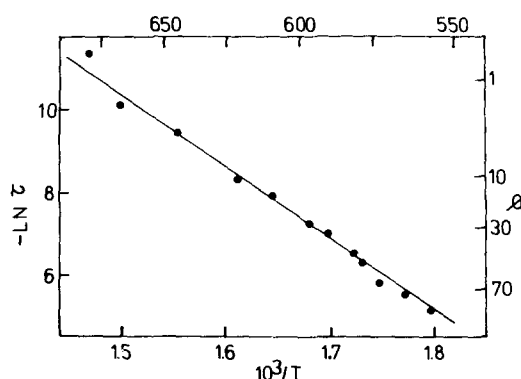


FIG. 10. The relaxation time τ obtained from the phase lag ϕ as a function of $1/T$ for a modulated CO beam. $\omega = 779 \text{ sec}^{-1}$.

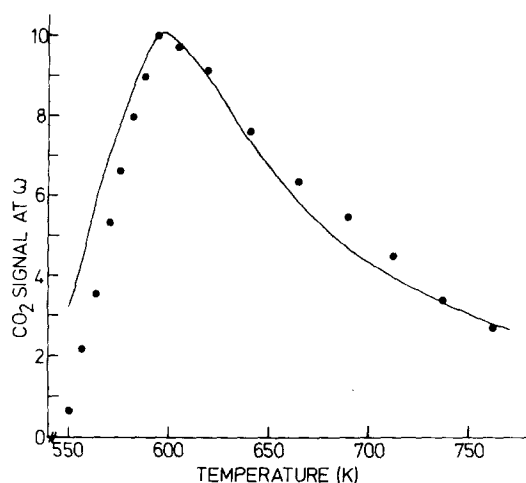


FIG. 11. Rate of CO_2 production as a function of temperature obtained with phase sensitive detection for a CO beam. $\omega = 779 \text{ sec}^{-1}$.

itself cannot be determined from the temperature dependence of the phase lag. This can, however, be achieved by fitting the experimentally observed variation of the CO_2 signal amplitude, J , with temperature. This quantity is, for the LH mechanism, contained in the second term within the brackets of Eq. (17) which simplifies for $k_4 f(\theta_0) \ll k_2$, and assuming θ_0 as well as k_1 ($=$ rate constant for CO adsorption) to be independent of temperature to

$$J = \text{const} \frac{k_4}{(k_2^2 + \omega^2)^{1/2}} \quad (20)$$

With $k_4 = \nu_4 \exp(-E_{\text{LH}}^*/RT)$ and the value of k_2 from Ref. 25 the experimental data (full circles in Fig. 11) can be fitted well with Eq. (20) if $E_{\text{LH}}^* = 25 \pm 2 \text{ kcal/mole}$. This value is in good agreement with the activation energy evaluated from the nonstationary measurements described in Sec. III. B and is valid for small surface concentrations of CO and $\theta_0 > 0.1$.

Obviously no such fit can be obtained by the use of Eq. (19), which again rules out the ER mechanism.

Similar measurements to those just discussed were also carried out with a modulated O_2 beam and a constant CO flux from the background atmosphere. As was outlined in Secs. III. A and III. B for $T > 500 \text{ K}$ and very low oxygen coverages the rate of CO_2 formation follows first-order kinetics with respect to the oxygen coverage, i. e., it may be formulated as

$$d\text{CO}_2/dt = k_4 \theta_0 \theta_{\text{CO}} \quad (\text{LH}), \quad (21)$$

and

$$d\text{CO}_2/dt = k_5 \theta_0 p_{\text{CO}} \quad (\text{ER}), \quad (22)$$

respectively.

Under experimental conditions for which $\theta_{\text{CO}} \gg \theta_0$ and $d\theta_{\text{CO}}/dt \approx 0$ the second-order kinetics simplify into a pseudo-first-order reaction. The justification of this assumption under the chosen experimental conditions will be discussed below. Substituting

$$p_{\text{O}_2} = p_{\text{O}_2}^0 + \alpha e^{i\omega t} \quad (23)$$

and

$$\theta_{\text{O}} = \theta_{\text{O}}^0 + \beta e^{i\omega t} \quad (24)$$

into Eqs. (21) and (22) then yields

$$\frac{d\text{CO}_2}{dt} = k_4 \theta_{\text{CO}} \left(\frac{\theta_{\text{O}}^0 + 2k_3 \alpha e^{i(\omega t - \phi)}}{(k_4^2 \theta_{\text{CO}}^2 + \omega^2)^{1/2}} \right) \quad (25)$$

with

$$\tan \phi_{\text{LH}} = \frac{\omega}{k_4 \theta_{\text{CO}}} \quad (26)$$

for the LH mechanism, and

$$\frac{d\text{CO}_2}{dt} = k_5 p_{\text{CO}} \left(\frac{\theta_{\text{O}}^0 + 2k_3 \alpha e^{i(\omega t - \phi)}}{(k_5^2 p_{\text{CO}}^2 + \omega^2)^{1/2}} \right) \quad (27)$$

with

$$\tan \phi_{\text{ER}} = \frac{\omega}{k_5 p_{\text{CO}}} \quad (28)$$

for the ER mechanism.

As already outlined, at $T > 500 \text{ K}$ and low CO pressure the reaction rate is directly proportional to the CO coverage, which may be expressed in terms of the rates for adsorption and desorption determining the equilibrium, viz.,

$$\theta_{\text{CO}} = \gamma \frac{p_{\text{CO}} S(\theta)}{\nu_2} e^{E_{\text{des}}^*/RT} \quad (29)$$

The rate of CO desorption is given by $\nu_2 e^{-E_{\text{des}}^*/RT} \theta_{\text{CO}}$; that for adsorption is proportional to $S(\theta) p_{\text{CO}}$, where $S(\theta)$ is the sticking coefficient. γ is the proportionality constant relating the CO pressure with impingement rate.

The temperature dependence of the phase shifts becomes evident by reformulating Eqs. (26) and (28) into

$$\tan \phi_{\text{LH}} = \frac{\omega \nu_2}{\gamma S(\theta) p_{\text{CO}} \nu_4} \exp(E_{\text{LH}}^* - E_{\text{des}}^*)/RT \quad (30)$$

and

$$\tan \phi_{\text{ER}} = \frac{\omega}{p_{\text{CO}} \nu_5} \exp(E_{\text{ER}}^*/RT), \quad (31)$$

where E_{LH}^* and E_{ER}^* are the activation energies governing the rate constants k_4 and k_5 , respectively, and ν_4 and ν_5 the corresponding pre-exponentials.

Since $E_{\text{des}}^* > E_{\text{LH}}^*$, which is suggested from the analysis of the preceding data ($E_{\text{des}}^* \approx 34 \text{ kcal/mole}$; $E_{\text{LH}}^* \approx 25 \text{ kcal/mole}$), a clear qualitative distinction can be made between both reaction mechanisms: For the LH mechanism the phase lag is predicted to increase from 0 to $\pi/2$ with increasing temperature, whereas with the ER mechanism it should decrease from $\pi/2$ to 0. The results shown in Fig. 12 are in clear agreement with the LH mechanism. The resulting apparent activation energy of -9.5 kcal/mole should, according to Eq. (30), be equal to $E_{\text{LH}}^* - E_{\text{des}}^*$. With $E_{\text{des}}^* = 34 \text{ kcal/mole}$ (the value of the isosteric heat of CO adsorption),²⁷ $E_{\text{LH}}^* = 24.5 \text{ kcal/mole}$ results, which is within the limits of error in good agreement with the values of 27 kcal/mole obtained with the nonstationary measurements and

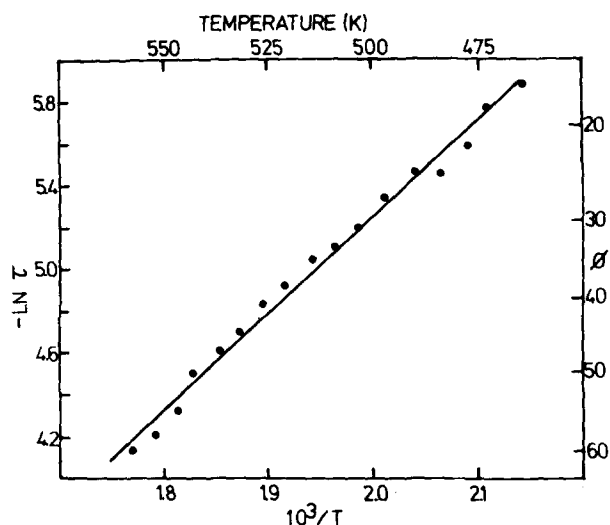


FIG. 12. The relaxation time τ obtained from the phase lag ϕ as a function of $1/T$ for a modulated O_2 beam. $\omega = 113 \text{ sec}^{-1}$.

of 25 kcal/mole resulting from the analysis of the data obtained with a modulated CO beam.

The variation with temperature of the amplitude of the rate of CO_2 formation by applying a modulated CO beam is reproduced in Fig. 13. For the LH mechanism, Eq. (25) predicts at first an increase proportional to $\exp(E_{LH}^*/RT)$, and then after passing a maximum a decrease proportional to $\exp(-E_{LH}^*/RT)$. For the ER mechanism, on the other hand, the amplitude should continuously rise with temperature as $\exp(-E_{ER}^*/RT)$ before asymptotically approaching a constant value at high temperatures. The data of Fig. 14 are in clear qualitative agreement with the behavior predicted for the LH mechanism, and in particular the decrease of the amplitude with increasing temperature rules out the operation of the ER mechanism. Quantitative agreement is, however, not achieved since the approximations underlying the derivation of Eq. (25) are no longer fulfilled over the whole temperature range: The amplitude

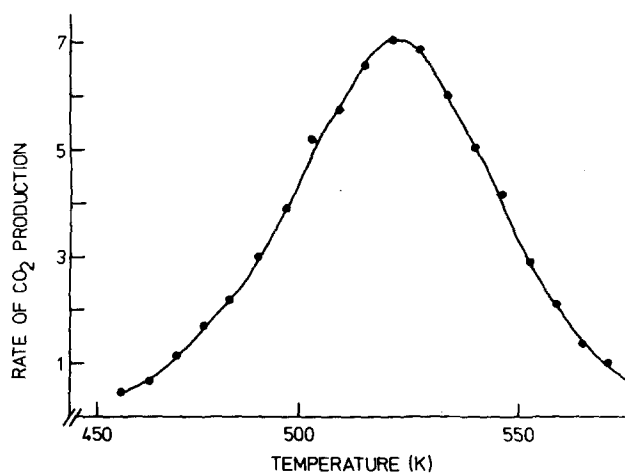


FIG. 13. Rate of CO_2 production as a function of temperature obtained with phase sensitive detection for an O_2 beam. $\omega = 113 \text{ sec}^{-1}$.

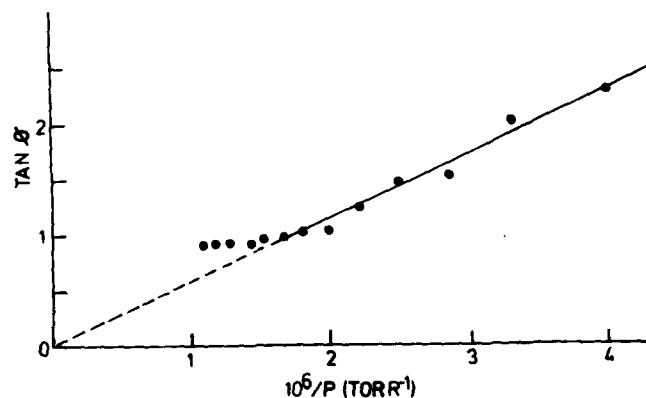


FIG. 14. $\tan \phi$ vs $1/p_{CO}$ for a modulated O_2 beam. $T = 505 \text{ K}$.

drops more rapidly from its maximum to both higher and lower temperatures than predicted by Eq. (25). The deviation at temperatures below the maximum is due to the increasing CO coverage which leads to a reduction of the oxygen sticking coefficient and thereby also of the rate of CO_2 formation. For temperatures above the maximum the exponential falloff in the CO coverage with increasing temperature leads to an increase in the oxygen coverage which in turn reduces the CO coverage by reaction even more. This results in a more rapid decrease of the amplitude than predicted by Eq. (25). According to Eq. (26), $\tan \phi$ should vary inversely with θ_{CO} at constant temperature. Since at the chosen temperature (505 K) θ_{CO} is proportional to p_{CO} , a plot of $\tan \phi$ vs $1/p_{CO}$ should therefore yield a straight line which passes through the origin. As can be seen from Fig. 14, this is indeed the case; the derivation from linearity at high CO pressures is accompanied by a decrease in reaction probability and may be due to the change in surface configuration at high CO coverages. According to Eq. (28) the linear relationship should equally be fulfilled with the ER mechanism so that this result yields no additional confirmation of the reaction mechanism. On the other hand, θ_{CO} can be derived from p_{CO} by the use of Eq. (29), and therefore from data of Fig. 14 the absolute value of the rate constant k_4 may be determined.

With the experimentally determined activation energy $E_{LH}^* = 25 \text{ kcal/mole}$ the rate constant can be written in the form $k_4 = 1.1 \times 10^{-12} \exp(-25000/RT)$ (particles sec^{-1}). The thus derived rate constant calculated for 387 K is a factor of 1000 smaller than the maximum value for k' shown in Fig. 6. This is reasonable since under the conditions shown in Fig. 6 the activation energy is considerably reduced by the compression of the oxygen adlayer. On the basis of this number the approximation underlying the analysis of the present data, viz., $k_4 \theta_0 \ll k_2$, may be re-examined. For $T > 500 \text{ K}$, $k_2 > 5k_4 \theta_0$, justifying the approximation made.

The dependence of $\tan \phi$ on θ_{CO} also allows an assessment to be made on how well the reaction approaches pseudo-first-order kinetics. An increase in the O_2 beam intensity at constant CO pressure and temperature should lead to a change in $\tan \phi$ if θ_{CO} is depleted by reaction. However, a variation of more than a factor of

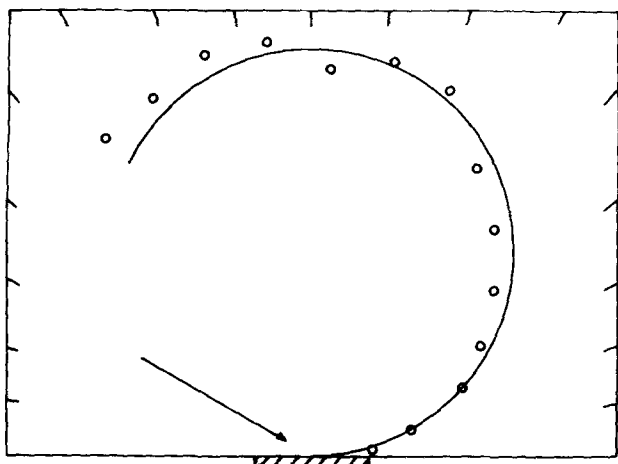


FIG. 15. Angular distribution for CO_2 emitted from the surface. A mixed beam of CO and O_2 was incident on the surface at the angle shown by the arrow. $T = 570 \text{ K}$, $\omega = 113 \text{ sec}^{-1}$.

5 in the O_2 beam intensity left the measured phase lag ϕ unchanged, justifying the assumption made.

D. Angular distribution of the CO_2 molecules formed

Figure 15 shows the angular distribution of the CO_2 molecules in the scattering plane at 570 K for a CO beam impinging on the surface which is immersed in an O_2 pressure of 1×10^{-7} torr. As is seen, the data can be fitted to a good approximation by a cosine distribution. This suggests that either both reactants or the CO_2 molecules are present on the surface for a time τ long enough that a preference for the specular direction is not pronounced. This result is consistent with, but need not imply, the L-H mechanism.

Although the angular distribution of the CO_2 formed has not been measured previously on Pd, results have been reported for the same reaction on Pt surfaces.^{14,15} Palmer and Smith¹⁴ observed an angular density distribution on Pt(111) which could be fitted with a function $\cos^d \alpha$ with d between 4 and 6. Becker *et al.*¹⁵ have reported an angular flux distribution with d between 2 and 3 for reaction on a polycrystalline Pt surface. Moreover, the velocity distribution of the CO_2 molecules was measured directly and it was found that those emitted perpendicular to the surface had a most probable velocity which was 30% greater than that of the CO_2 emitted at 45 deg relative to the surface normal.

Since the detector used in the present work measures the density distribution, the data must be corrected by a factor $1/v(\alpha)$, where $v(\alpha)$ is the velocity of the CO_2 molecules at the angle α , to obtain the flux distribution. If it is assumed that the same velocity anisotropy is present in this work as was reported for polycrystalline Pt,¹⁵ the angular flux distribution in this study would be described by a d value between 1 and 2. However, the same correction must be applied to the work of Palmer and Smith,¹⁴ which would increase their d value even more.

A more detailed discussion of the angular distribution in flux and velocity and the theoretical models used in

their explanation^{29,30} must await further measurements on single crystal surfaces. As yet, only one measurement on single crystal surfaces which have been checked *in situ* to be clean has shown a deviation from the cosine and Maxwell-Boltzmann distributions.²⁹ For this system— H_2 on Cu—the adsorption is known to be activated³¹ and the results can be explained in terms of a barrier model.³² However, the application of a similar model to CO oxidation is questionable since there is no reason to believe that the adsorption of CO_2 is an activated process.

IV. DISCUSSION

A. Reaction mechanism

Due to the existence of a series of rather detailed previous studies on the interaction of CO and O_2 with palladium surfaces most of the possible reaction steps as outlined by Eqs. (11) and (12) are well established and others one could think of can clearly be ruled out.

(i) On a clean Pd(111) surface CO adsorbs nondissociatively²⁴ with a high sticking probability ($S_{\text{O}} = 0.96$ ²⁵) and an adsorption energy of 34 kcal/mole.²⁷ Above 450 K the mean surface residence time drops below 10^{-2} sec so that the desorption (k_2) has to be taken into account and is of particular importance for the kinetics of CO_2 formation under steady-state conditions. Several ordered overlayer structures are formed with increasing coverage. Saturation is characterized by the formation of a hexagonal densely packed layer at $\theta = 0.66$ ($\approx 1 \times 10^{15}$ molecules/cm²).²⁴

(ii) O_2 adsorption on a clean Pd(111) surface takes place dissociatively and without any noticeable activation energy. Saturation of the O adlayer is characterized by the formation of a 2×2 structure ($\theta_{\text{O}} = 0.25$), which, however, may be "compressed" into a $\sqrt{3} \times \sqrt{3}/R$ 30° structure (local coverage $\theta_{\text{O}} = 0.33$) by the presence of small amounts of coadsorbed CO.^{24,26} The initial sticking coefficient is 0.4²⁵ and the heat of adsorption about 55 kcal/mole.²⁶ Desorption and bulk diffusion attain significant rates only at temperatures above 700 K, so that these processes play no role under the present conditions and only k_3 ($\hat{=}$ rate of oxygen adsorption) has to be considered with Eqs. (11b) and (12a).

(iii) Adsorption of both species on the surface is rather complicated, or in other words the rate constants k_1 – k_3 are functions of both θ_{O} and θ_{CO} as well as of temperature²⁴: k_3 (i.e., the sticking coefficient for oxygen) drops strongly with increasing CO coverage. For $\theta_{\text{CO}} \geq 0.33$ adsorption of O_2 is completely inhibited. No such inhibition effect is observed for CO adsorption on a surface precovered with oxygen. That means that even if the surface is saturated with O_{ad} ($\theta = 0.25$) appreciable amounts of CO (up to a total CO coverage of about 0.5) may still be adsorbed. If CO is adsorbed first to $\theta_{\text{CO}} < 0.33$ additional uptake of oxygen leads to the formation of separate islands of CO_{ad} and O_{ad} . If, on the other hand, CO is admitted to a O-covered surface ($\theta_{\text{O}} = 0.25$) the O- 2×2 structure is at first compressed into domains with local oxygen coverage 0.33 ($\sqrt{3} \times \sqrt{3}/R$ 30° structure) and CO_{ad} forms separate domains. At still

higher CO exposures a mixed phase (2×1 structure) is formed where the unit cell contains both species in intimate contact. These processes not only affect the mutual configurations of the reactants but also their effective adsorption energy and the activation energy for CO_2 formation, i.e., k_4 of Eq. (11c).²⁴

These briefly summarized effects will be of essential importance for the reaction kinetics which will be discussed in the next section. Before this can be done the vital question, however, is concerned with the mechanism of the surface reaction itself, i.e., whether this proceeds via $\text{CO}_{\text{ad}} + \text{O}_{\text{ad}} \rightarrow \text{CO}_2$ [Eq. (11c)] (LH mechanism) or by the collisiontype ER mechanism $\text{CO} + \text{O}_{\text{ad}} \rightarrow \text{CO}_2$ [Eq. (12b)]. Previously it has been suggested that the reaction proceeds predominantly via this latter path,^{14,16-18} mainly with the following arguments: (a) Exposure of an oxygen covered Pd(111) surface at room temperature to gaseous CO causes fairly rapid CO_2 formation, whereas separate domains of O_{ad} and CO_{ad} (as visible in the LEED pattern) are rather stable at this temperature. It was therefore concluded that a CO molecule striking O_{ad} from the gas phase reacts more easily to CO_2 than if it is in the chemisorbed state.¹⁶ A recent detailed low-temperature study showed clearly that this conclusion was wrong since different mutual arrangements of $\text{O}_{\text{ad}} + \text{CO}_{\text{ad}}$ may occur which also exhibit different tendencies for CO_2 formation.²⁴ The induction period observed in the nonstationary CO_2 production if a oxygen-covered surface is exposed to CO is a direct consequence of this effect and can never be explained on the basis of an ER mechanism. (b) The main argument in favor of the ER mechanism at $T > 450$ K was that the reaction rate decreases much less rapidly with increasing temperature than the equilibrium CO coverage. With the ER mechanism, on the other hand, the rate should be given by $r = k_5 \theta_{\text{O}} p_{\text{CO}}$. Since k_5 was assumed to be only slightly dependent on temperature (small activation energy) the in reality observed decrease of r with increasing T was attributed to a temperature dependent diffusion of oxygen into the bulk and at higher temperatures to the onset of O_2 desorption, thus effectively decreasing θ_{O} with increasing temperature. This study, however, has revealed that these processes are negligible for $T \leq 700$ K, at which temperature the stationary reaction rate has fallen to about 10% of its maximum value. With the LH mechanism the rate should decrease at high temperatures as $\exp(E_{\text{ad}}^{\text{CO}} - E_{\text{LH}}^*)/RT$ rather than with $\exp(E_{\text{ad}}^{\text{CO}}/RT)$ which is in agreement with the present experimental data. Thus the already mentioned arguments which are based on more recent results clearly favor the operation of the LH mechanism even at higher temperatures.

Additional arguments supporting this suggestion are obtained from the present relaxation type measurements. The difference between both mechanisms is most clearly seen in the temperature dependence of the phase lag ϕ in Figs. 10 and 12. Analysis of the data as outlined in Sec. III.C reveals that the temperature dependence of the phase shift as observed with a modulated CO beam clearly rules out the ER mechanism. Similarly, the observation that $d\phi/dT > 0$ for a modulated O_2 beam can only be reconciled with the LH mechanism and is in

contradiction to the prediction of the ER model. The good agreement between the reaction activation energies derived on the basis of the LH model from the modulated beam experiments and from the nonstationary measurements is further support for this mechanism. It may be concluded from these results that a unequivocal distinction between both types of reaction mechanisms discussed may only be achieved on the basis of relaxation experiments.

B. Reaction kinetics

Rate laws in kinetics are usually formulated as a function of the partial pressures of the species involved in the reaction. In the present case one would consequently be interested in an expression of the form $r = d\text{CO}_2/dt = f(T, p_{\text{CO}}, p_{\text{O}_2}, p_{\text{CO}_2})$ ³² describing the kinetics of the overall reaction $2\text{CO} + \text{O}_2 \rightarrow 2\text{CO}_2$. Usually experimental data are fitted to empirical rate laws, for example by varying the reaction orders. In heterogeneous catalysis this procedure will generally yield only very little information on the underlying mechanism of the reaction—in contrast to what is sometimes believed. On the other hand, a reaction mechanism, once it is established in detail by independent pieces of information, should allow the formulation of a theoretical overall rate expression which agrees with the experimental data.

In the present case the sequence of reaction steps



looks rather simple, and therefore at a first glance a straightforward derivation of the rate law appears to be a fairly easy task. This is, however, by no means the case, since the rate "constants" k_1 – k_4 depend not only on temperature but in a rather complex manner also on the concentrations and mutual configurations of the reacting species on the surface. In fact, only systems of independent noninteracting particles in a random arrangement (such as present in normal gas-phase kinetics, but also underlying the model of the Langmuir adsorption isotherm) will exhibit rate constants which are only functions of the temperature. (More exactly, even then at least the concentrations of occupied and unoccupied adsorption sites on the surface have to be taken additionally into account. The fact that different kinds of particles will also usually occupy different numbers of surface atoms as well as possible modifications of the adsorption kinetics by the existence of "precursor" states, etc., are then still neglected.) It appears to be rather questionable whether such a situation will ever be found with heterogeneous catalysis. In view of this situation any attempts to derive unambiguous information on the microscopic mechanism of a heterogeneously catalyzed reaction become even more speculative.

After these general critical remarks we return to the present work: Here the rate laws governing the individual reaction steps were at first determined experi-

mentally, and then they were used to establish the overall reaction mechanism as formulated in a rather simplified manner by Eq. (11). Now this mechanism together with the information on the individual rate laws will be used in an attempt to explain the various phenomena observed with the kinetics of CO_2 formation. In order to rationalize the discussion the procedure will be split into two parts: According to Eq. (11c) the rate of CO_2 formation is obviously a function of T , θ_{O} , and θ_{CO} . However, it may not be simply written as $r = k_4 \theta_{\text{O}} \theta_{\text{CO}}$ as suggested by the stoichiometry of this reaction step. In general, O_{ad} and CO_{ad} will not be randomly distributed over the surface and in contrast the particular configuration of the adsorbed species as mentioned in Sec. IV.A and studied in detail in Ref. 24 will be responsible for the corresponding kinetics of Reaction (11c). The second part then consists in correlating θ_{O} and θ_{CO} under the corresponding conditions of the reaction with the partial pressures p_{O_2} and p_{CO} . Again no simple relations as suggested by Eqs. (11a) and (11b) hold, for various reasons. As a consequence no analytical expression for Eq. (32), i.e., the rate law of the overall reaction, but rather several approximations for limiting cases as they were already used in the previous sections for analysing the experimental results may be presented. Throughout the whole discussion the partial pressure and coverage of CO_2 will never come into play: Since this molecule is only very weakly held on the surface any noticeable coverages will never be built up under the applied conditions. In addition, decomposition of CO_2 was never observed to occur on Pd surfaces.

1. Low surface concentrations of O_{ad} and CO_{ad}

If θ_{O} and θ_{CO} are far below saturation, the adsorbed particles may safely be considered to form a random configuration (i.e., there is no indication for condensation into isolated domains)²⁴ and the rate of CO_2 formation is expected to follow simple second order kinetics, viz.,

$$r = k_4 \theta_{\text{O}} \theta_{\text{CO}} \quad (32)$$

This situation will be achieved during the steady-state reaction if the following conditions are fulfilled:

(i) The temperature must be high enough that the equilibrium $\text{CO} \rightleftharpoons \text{CO}_{\text{ad}}$ is shifted to the left-hand side. According to the equilibrium isotherms of this system²⁷ this will be the case for $T \geq 500$ K for $p_{\text{CO}} \leq 10^{-6}$ torr, or above the temperatures at which the steady state-rate $r(T)$ reaches its maximum as shown in Fig. 2. The decrease in r with increasing T is then obviously due to the continuous decrease of θ_{CO} . At low CO coverage, $\theta_{\text{CO}} \propto p_{\text{CO}} \exp(E_{\text{ad}}^{\text{CO}}/RT)$, where $E_{\text{ad}}^{\text{CO}}$ is the initial adsorption energy for CO (34 kcal/mole). However, the apparent activation energy E_{app}^* derived from the high temperature data of Fig. 2 (a plot of $\ln r$ vs $1/T$) will not be equal to $(E_{\text{LH}}^* - E_{\text{ad}}^{\text{CO}})$ as one might assume since θ_{O} will also vary over the temperature range investigated owing to the temperature dependence of S_{react} . An apparent rate constant k'' is therefore introduced without further analysis.

(ii) p_{O_2} should be smaller than or equal to p_{CO} for the

following reasons: The sticking coefficients for $\theta \approx 0$ are $S_{\text{CO}} = 1$ and $S_{\text{O}_2} = 0.4$.²⁵ Since for O_2 adsorption two O adatoms are formed, there are roughly equal probabilities for the formation of CO_{ad} and O_{ad} at $p_{\text{O}_2} = p_{\text{CO}}$. If $p_{\text{CO}} > p_{\text{O}_2}$, each oxygen adatom should be immediately consumed by the reaction and the adsorption of O_2 becomes rate limiting, i.e., at constant temperature $r \propto p_{\text{O}_2}$. Inspection of Fig. 2 shows that this is indeed the case for Curve *a* in Fig. 2(a) and for Curves *a* and *b* in Fig. 2(b), where both conditions are fulfilled up to $p_{\text{O}_2} \approx p_{\text{CO}}$. Consequently, if $p_{\text{CO}} < p_{\text{O}_2}$, θ_{O} should continuously increase (since desorption is negligible) and under steady-state conditions the reaction proceeds on an oxygen saturated surface and the rate becomes independent of p_{O_2} as is observed in Curves *a* of Figs. 2(a) and 2(b).

Since under the conditions in (i) and (ii) the adsorption of O_2 will be rate limiting, Eq. (32) simply transforms into

$$r = 2k_3 p_{\text{O}_2} \quad (33)$$

where k_3 is the rate constant for O_2 adsorption.

For this case, r should be independent of p_{CO} which has been reported previously.¹⁴ Curves such as those shown in Fig. 2 for substrate temperatures greater than 640 K also show a similar behavior over a limited range of CO pressures.

The activation energy for the LH reaction under these conditions can best be determined from the modulated O_2 beam measurements for which $\theta_{\text{O}} < 0.1$ and was found to be 25 kcal/mole, which substantiates an earlier estimate²⁴ based on thermal desorption data. Because of the low coverages of both species, this activation energy obviously holds for a collision type mechanism between O_{ad} and CO_{ad} where the bond strengths of both adatoms are not noticeably affected by the presence of neighboring adatoms not directly involved in the reaction.

2. Low CO coverages, $\theta_{\text{O}} \geq 0.08$

These conditions are fulfilled for $T > 500$ K and $p_{\text{CO}} \leq 10^{-6}$ torr since for these temperatures and pressures, $\theta_{\text{CO}} \leq 0.02$, which is sufficiently low to prevent inhibition of oxygen adsorption. Figure 8 shows clearly that in this regime r is independent of the oxygen coverage. LEED investigations²⁴ show that under these conditions the oxygen adatoms form islands with a local coverage of $\theta = 0.25$ and the results are consistent with a reaction of CO_{ad} with O_{ad} at the island boundaries. The zero order kinetics with respect to oxygen can be understood on the basis of the following model.

Since the binding energy of oxygen on the surface is nearly twice that for CO, CO will diffuse over the surface at temperatures at which the oxygen is immobile and at these temperatures, the reaction takes place between a quasi-two-dimensional gas of CO and a fixed matrix of O_{ad} . As long as the mean CO diffusion distance is greater than the separation of oxygen islands, the rate should be independent of θ_{O} . Since the mean CO diffusion distance decreases exponentially with temperature, a departure from the zero order behavior seen in Fig. 7 should be observed at higher temperatures. This

is indeed the case; curves similar to those for r shown in Figs. 6 and 7 for $T > 650$ K show a nearly linear decrease of r with time. Diffusion is still important here since in the absence of diffusion, r should decrease exponentially with time. However, the trend from zero order to first order kinetics with respect to θ_O with increasing temperature predicted by the diffusion model is observed. The dependence of r on p_{CO} is linear in this entire regime, which does not imply an ER mechanism since θ_{CO} and p_{CO} are directly proportional at these temperatures and r is given by

$$r \propto k_4 \theta_{CO} \propto p_{CO} \nu_4 \exp(E_{CO}^{ad} - E_{LH}^*)/RT. \quad (34)$$

The activation energy for this coverage regime can be obtained from the modulated CO measurements (Figs. 10 and 11) and from the high temperature instationary results in Fig. 8. In both cases, a value of approximately 25 kcal/mole was obtained, which is identical to that obtained under the conditions described in Sec. IV.B.1. This is not surprising, since the elementary step is the same in both cases and the reaction rate constant rather than the diffusion coefficient is rate limiting. As long as the CO coverage is low enough that the $p(2 \times 2)$ oxygen structure is not compressed, the activation energy seems to be approximately 25 kcal/mole.

3. Moderate and high CO concentrations

If the temperature is low enough so that appreciable CO coverages may build up on the surface (i.e., $T < 500$ K under the present conditions), two new effects come into play: The first is concerned with the fact that adsorbed CO inhibits the dissociative adsorption of O and thereby also the degree of CO_2 formation. This effect limits the reaction rate under steady-state conditions, as will be discussed below. The second effect was already observed in previous nonstationary low-temperature experiments²⁴: An increase of the CO coverage of an O_{ad} -precovered surface ($\theta_O = 0.25$) by continuous admission on gaseous CO causes a compression of the oxygen adlayer into domains with a $\sqrt{3} \times \sqrt{3} R 30^\circ$ structure with a local O coverage $\theta_{O, local} = 0.33$. The adsorbed CO molecules occupy the thus created bare patches on the surface, again with a local CO coverage $\theta_{CO, local} = 0.33$. A simple calculation shows that the latter will consist of 25% of the total surface area, i.e., the total CO coverage will be $\theta_{CO, tot} = 0.08$. (If the initial O coverage is < 0.25 , a correspondingly larger total CO coverage will be needed in order to reach this type of domain formation.) With the closer O-O distance within the $O-\sqrt{3} \times \sqrt{3} R 30^\circ$ structure repulsive interactions between neighboring oxygen atoms come into play which lower their effective adsorption energy and also their activation energy for reaction with CO_{ad} . The latter was roughly estimated in our previous TDS study to be about 18 kcal/mole. As a consequence, CO_2 formation will start already at lower temperatures than before [respectively the rate at constant (low) temperature will be increased].

These two phenomena will now serve to explain the observed kinetic effects: The Arrhenius plot of Fig. 8 shows a smaller slope at temperatures below 500 K.

In this temperature range $\theta_O = 0.25$ and $\theta_{CO} \leq 0.06$, i.e., this situation corresponds to the just discussed compression of the O adlayer.

The steady-state rate of CO_2 formation as reproduced in Fig. 1 exhibits a continuous increase up to a maximum value as observed several times in earlier studies. Previous measurements showed that in this range the CO coverage under reaction conditions continuously decreases due to increasing CO desorption. As a consequence the sticking coefficient for O_2 continuously increases. To a first approximation the rate law may be expressed as $r = k''' p_{O_2}/p_{CO}$,³³ where the apparent activation energy is roughly equal to the adsorption energy of CO. The maximum rate shifts towards higher temperatures with increasing p_{CO} for obvious reasons since now higher temperatures are needed in order to decrease the CO coverage which inhibits oxygen adsorption. Since at higher temperatures and equal θ_{CO} and θ_O on the other hand the reaction probability (due to the existence of an activation energy for the LH reaction) becomes higher, the maximum rate also increases.

The activation energy decreases from 25 kcal/mole to 14 kcal/mole through the change in the geometry of the adlayer. This result substantiates the estimate made on the basis of flash desorption curves.²⁴

In Fig. 2 the curves taken at higher CO pressures can also be explained on the basis of the inhibition effect of adsorbed CO: At constant T and p_{O_2} the steady-state rate decreases with increasing p_{CO} . At constant p_{CO} with increasing p_{O_2} , however, more of the adsorbed CO molecules are consumed by the LH reaction, thus enabling a stronger than linear increase of θ_O with p_{O_2} . Consequently, the plateau at which the surface is saturated with O_{ad} is also reached only at higher O_2 pressures. The parabolic shape of these curves suggests a rate law of the form $r \propto p_{O_2}^2$ which, of course, has nothing to do with a mechanism which is of true second order kinetics with respect to oxygen; it merely reflects the dependence of the sticking coefficient of O_2 on the CO coverage.

The nonstationary measurements reproduced in Fig. 3, where an oxygen saturated surface was suddenly exposed to a CO beam, exhibit at first an induction period in the rate of CO_2 formation. The origin of this effect (which could hardly be understood with an ER mechanism) is caused by the initial compression of the O adlayer and not by an initially smaller rate of CO adsorption as one might suspect. This becomes evident from an inspection of Fig. 6, where a similar experiment performed at 374 K (in contrast to Fig. 3, however, not with a steady-state O_2 atmosphere but only with an O-precovered surface) was analyzed in detail: In fact the CO coverage rises rather rapidly during the first second, during which r increases only slowly and θ_O remains almost constant. Appreciable consumption of O_{ad} starts only after about 1 sec when on the other hand the CO coverage has attained a value of $\theta_{CO} \approx 0.1$. During the course of CO_2 formation additional CO has to be adsorbed in order to refill that portion of the total surface area which is depleted in O_{ad} through reaction and therefore the maximum rate is only slowly attained.

The situation is completely different if the same experiment is performed at 509 K (Fig. 7): Now θ_{CO} remains always very small and the compression effect becomes insignificant. However, now the LH reaction with normal O_{ad} ($E^* = 25$ kcal/mole) dominates and therefore the maximum rate is observed immediately after switching on the CO pressure.

The shape of the curves in Fig. 3 for $T < 500$ K may be explained as follows: After the induction period the maximum rate is observed at a constant level for a relatively long time. Nevertheless, the oxygen coverage even in an O_2 atmosphere continuously decreases since CO striking the surface competes with O_2 for the sites depleted in O_{ad} by the reaction. Since, however, the rate is independent of the oxygen coverage for $\theta_{\text{O}} > 0.1$, as outlined above, it remains constant until the CO coverage is so high that θ_{O} drops below this critical value. As a consequence the rate then decreases to its steady-state value determined by the stationary oxygen coverage which is strongly suppressed by the presence of adsorbed CO. At 570 K the CO coverage is negligibly small and no inhibition of oxygen adsorption takes place. Therefore, r retains its constant value.

The remaining kinetic results were already analyzed in connection with the corresponding measurements so that their discussion will not be repeated. In conclusion, all observations fit well into the established reaction scheme of Eq. (11) which, however, in spite of its simplicity turned out to exhibit rather complex kinetics for the outlined reasons.

V. CONCLUSIONS

Both the stationary and nonstationary measurements show that the CO oxidation reaction takes place between two adsorbed species (LH mechanism); no evidence for an ER mechanism was found. However, in general the rate of CO_2 formation cannot be written as the product of a bimolecular rate constant and the coverages of the two adsorbed species for the following reasons:

(i) CO adsorption reduces the sticking coefficient for O_2 if $\theta_{\text{CO}} > 0.06$. Therefore at higher coverages $r \propto 1/\rho_{\text{CO}}$.

(ii) At intermediate oxygen coverages ($0.08 < \theta_{\text{O}} < 0.25$) and for $T > 500$ K, the reaction takes place between rapidly diffusing CO molecules and stationary oxygen islands. In this regime r is independent of θ_{O} .

(iii) At moderate CO and oxygen coverages, the oxygen is compressed from a $p(2 \times 2)$ to a $\sqrt{3} \times \sqrt{3} R30^\circ$ structure which results in a reduction of the activation energy for reaction from 25 kcal/mole to 14 kcal/mole and causes a complex dependence of r on θ_{O} and θ_{CO} .

The simple rate expression was found to apply only in the special case of low θ_{O} and θ_{CO} . Although the LH mechanism applies throughout the temperature and pressure ranges investigated, a quantitative description of the kinetics must consider factors such as the changes in sticking coefficients due to coadsorption and their temperature dependence, the relative rates of dif-

fusion of the two adspecies and the surface configuration as a function of temperature and pressure.

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